No. of the Experiment: ………...

Name of the Experiment: Determination of the acceleration due to gravity “g” at a place using compound pendulum.

Theory:

When a body is left to free fall, then it acquires a constant acceleration and move towards earth, due to earth’s gravitation. Now the constant acceleration acquired by that freely falling body is called acceleration due to gravity.

Time period of oscillation of a physical pendulum or compound pendulum is given by

$$T = 2\pi\sqrt{\frac{K^2+D^2}{gD}} \quad (Sec)$$

Compare the above formula with time period of oscillation of a simple pendulum. i.e.

$$T = 2\pi\sqrt{\frac{L}{g}} \qquad (Sec)$$

Here, $L = \frac{K^2+D^2}{D}$ is called length of equivalent simple pendulum. K is radius of gyration; ‘D’ is distance of the point of suspension from center of gravity. The value of “L” is estimated from a graph drawn between distance of point of suspension (l) verses time period of oscillation (T)

From the above formula acceleration due to gravity is given by

$$g = 4\pi^2\left(\frac{L}{T^2}\right) \qquad (cm/s^2)$$

Derivation:

In lower classes you might be familiar with simple pendulum experiment for the determination of acceleration due to gravity, hence before to do this experiment one has to know the difference between simple pendulum, and compound pendulum experiment, both of them were meant for determination of ‘g’ value. They are

1. Simple pendulum is an ideal case, because it requires a point mass object
2. It requires torsion less string.

The above mentioned two conditions are not practically possible, hence it is only a mathematical ideal case, and whereas compound pendulum is a physical pendulum. A rigid body of any shape which is free to oscillate without any friction on a vertical plane is called compound pendulum. It swings harmonically back and forth about a vertical z-axis (Passing through point “O” as shown in Fig), when compound pendulum is displaced from its equilibrium position by an angle θ. In the equilibrium position, the center of gravity of the body is vertically below at a distance of OG. Let the mass of the body is m, in this experiment you are going to measure the acceleration due to gravity, g by observing the motion of a compound pendulum. Let us consider a compound pendulum shown in Figure 1.

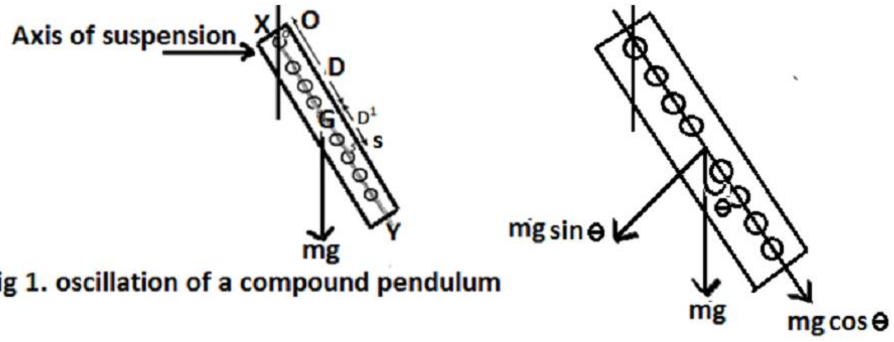


Fig 1. oscillation of a compound pendulum

Pull the compound pendulum through an angle θ and release it, then it makes angular oscillations due to torque acting on it, given by

$$\vec{\tau} = \vec{r} \times \vec{F}$$

=> $$\vec{\tau} = OG \times mgsin\theta$$

=> $$\vec{\tau} = -mgDsin\theta \text{ -----------------------------(1)}$$

Here –ve sign is because of force and displacement is opposite to each other.

For small amplitudes, $sin\theta \approx \theta$

Now expression (1) becomes, $\vec{\tau} = -mgD\theta$

We know that torque, $\vec{\tau} = I\alpha$

(Here, $I = Moment\ of\ Inertia$ about an axis passing through point "O". and $\alpha = Angular\ Acceleration$)

=> $$-mgD\theta = I\frac{d^2\theta}{dt^2}$$

=> $$\frac{d^2\theta}{dt^2} + \frac{mgD}{I}\theta = 0 \text{ ----------------------(2)}$$

Equation (2) represents simple harmonic equation of the form, i.e.

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

Here, $\omega =$ is angular frequency of simple pendulum. From comparison with Eq (2), we can write

$$\omega = \sqrt{\frac{mgD}{I}}$$

=> $$\omega = \frac{2\pi}{T} = \sqrt{\frac{mgD}{I}}$$

$$T = 2\pi\sqrt{\frac{I}{mgD}} \text{ --------------------------(3)}$$

According to parallel axes theorem, the rotational moment of inertia, about any axis parallel to the one passing the center of gravity is given by

$$I = I_G + mD^2 \text{ -------------------------(4)}$$

We know that moment of inertia about an axis passing through center of gravity "G", given by

$$I_G = mK^2$$

Here "K" is radius of gyration of the body about an axis passing through "G".

Thus, $$T = 2\pi\sqrt{\frac{mK^2 + mD^2}{mgD}} = 2\pi\sqrt{\frac{D + \frac{K^2}{D}}{g}} \text{ ---------------(5)}$$

Comparing expression (5) with expression for time period of simple pendulum i.e.

$$T = 2\pi\sqrt{\frac{L}{g}}$$

This suggests that, $L = D + \frac{K^2}{D} = D + D_1$ -----------------------(6) (Let, $D_1 = \frac{K^2}{D}$)

The term “L” is called length of “equivalent simple pendulum”. This is because simple pendulum of length “L” is having a time period, same as that of time period of compound pendulum. Also, it seems that all the mass of the body were concentrated at point “S”, along “OG” produced such that

$$OS = D + \frac{K^2}{D} = D + D_1$$

The point “S” is called center of oscillation. In analogy with simple pendulum, we may suppose that, the entire mass is concentrated at that point.

From expression (6), the extra distance $D_1 = \frac{K^2}{D}$ is below the center of gravity “G”, at a point “S”, and is shown in Figure (1).

From expression (6) we can write,

$$D^2 - LD + K^2 = 0$$

The above equation is a quadratic equation in “D” and it’s two roots are given by

$$D_1 = \frac{L+\sqrt{L^2-4K^2}}{2} \quad \text{and} \quad D_2 = \frac{L-\sqrt{L^2-4K^2}}{2}$$

That is for each half of pendulum, there are two different points of oscillation (do not get confusion with center of oscillation) i.e., which are at D_1 and D_2 distance away from center of gravity “G”, for which the value of “L” is same. Since “L” is same for D_1 and D_2, then, the time period is also same. When we perform this experiment on both sides of center of gravity “G” we have a total of 4 points (2 points on one side) having same time period “T”, as shown in Figure 2. The points D and D_2 are clearly shown in Graph.

It is sometimes convenient to specify, the location of axis of suspension or point of oscillation “O”, by the distance from end of the bar, instead of distance “D” from center of gravity. By varying the position of axis of suspension, measure the corresponding time period, and tabulate all the observation in the table.

List of the Apparatus:

1. Compound pendulum
2. Knife edges
3. Stop watch
4. Meter scale

Procedure:

1. The compound bar pendulum AB is suspended by passing a knife edge through the first hole at the end A.
2. The pendulum is pulled aside through a small angle (Max. 4^0) and released, whereupon it oscillates in a vertical plane with a small amplitude.
3. The time for 20 oscillations is measured. From this the period *T* of oscillation of the pendulum is determined.
4. In a similar manner, periods of oscillation are determined by suspending the pendulum through the remaining holes on the same side of the centre of mass G of the bar.
5. The bar is then inverted and periods of oscillation are determined by suspending the pendulum through all the holes on the opposite side of G.
6. The distances *d* of the top edges of different holes from the centre of gravity, G of the bar are measured for each hole.

7. The position of the centre of mass of the bar is found by balancing the bar horizontally on a knife edge.
8. A graph is drawn with the distance *d* of the various holes from the end A along the X-axis and the period *T* of the pendulum at these holes along the Y-axis. The graph has two branches, which are symmetrical about G.
9. To determine the length of the equivalent simple pendulum corresponding to any period, a straight line is drawn parallel to the X- axis from a given period *T* on the Y- axis, cutting the graph at four points A, B, C, D.
10. The distances AC and BD, determined from the graph, are equal to the corresponding length *l*. The average length *l* = (AC+BD)/2 and l/T^2 are calculated.
11. In a similar way, l/T^2 is calculated for different periods by drawing lines parallel to the X-axis from the corresponding values of *T* along the Y- axis. l/T^2 should be constant over all periods *T*, so the average over all suspension points is taken. Finally, the acceleration due to gravity is calculated from the equation $g = 4\pi^2(l/T^2)$.

Observations:

Table: For the determination of time periods.

Sl No.	No. of Holes	Distance of the Holes from Center of Gravity, G (cm)	On one side of Center of Gravity, G (Knife edge A)				On other side of Center of Gravity, G (Knife edge B)			
			Time for 20 vibrations			Time periods, $T = \frac{t}{20}$ (sec)	Time for 20 vibrations			Time periods, $T = \frac{t}{20}$ (sec)
			t_1 (sec)	t_2 (sec)	Mean, $t = \frac{t_1 + t_2}{2}$ (sec)		t_1 (sec)	t_2 (sec)	Mean, $t = \frac{t_1 + t_2}{2}$ (sec)	
1.	4									
2.	3									
3.	2									
4.	1									

Calculations:

From graph, $T =$ ………….. (sec)

$AC =$ …………….. (cm)

$BD =$ ……………. (cm)

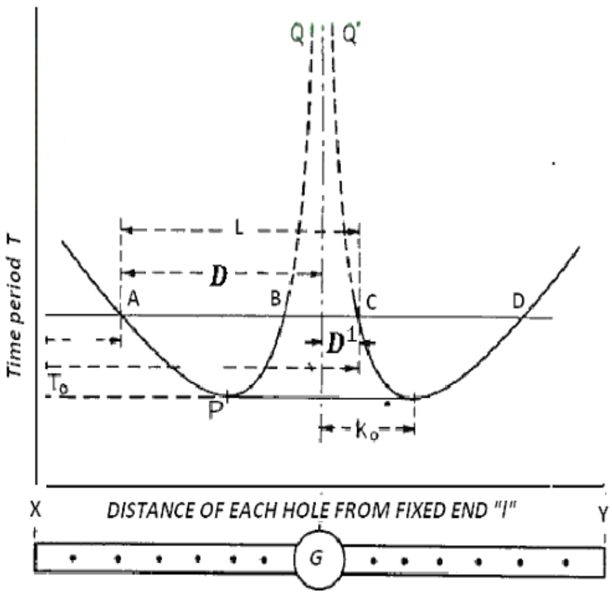
So, $L = \frac{AC+BD}{2} =$ ……………. (cm)

Now, the acceleration due to gravity is –

$$g = 4\pi^2 \left(\frac{L}{T^2}\right)$$

$$= \ldots\ldots\ldots \quad (cm/s^2)$$

$$= \ldots\ldots\ldots \quad (m/s^2)$$



Result:

Determination of the acceleration due to gravity "g" at this place = ……….. (m/s^2)

Precautions:

1. The knife edge is made horizontal.
2. If the knife edge is not perfectly horizontal the bar may be twisted while swinging.
3. The motion of bar should be strictly in a vertical plane.
4. Time period should be noted only after all types of irregular motions subside.
5. The amplitude of swing should be small (4º – 5º). So that the condition $sin\theta \approx \theta$ assumed in the derivation of formula remains valid.
6. The time period of oscillation should be determined by measuring time for a large number of oscillations with an accurate stop watch.
7. The graph should be drawn smoothly.
8. All distances should be measured and plotted from one end of the rod.